

Byzantine Accountability Without Consensus: Strong Eventual Consistency for Non-Associative, Stochastic, Robust Aggregation

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Abstract

Byzantine-robust aggregation rules such as multi-Krum assume a central coordinator, and decentralising them is obstructed by the rules themselves: they are globally coupled, non-associative, and discontinuous, so an ulp-scale perturbation can flip the selected subset, moving the output by a non-vanishing amount. None of this prevents coordinator-free replication, because a robust rule needs no agreed *order* of contributions, only an agreed *set* and an agreed *exclusion predicate*, both of which converge without consensus. ACFA (Accountable Consensus-Free Aggregation) replicates a content-addressed OR-Set of signed contributions and a grow-only set of self-authenticating equivocation proofs, offline-verifiable by anyone. Aggregation is a deterministic pure function of the converged product state: fixed-point integer arithmetic over a hash-canonical order, ties broken by content hash. We prove that any pure function of a converged product of CRDTs (non-monotone, non-associative, or stochastic) inherits Strong Eventual Consistency, together with its converse; the contribution is the composition of a data lattice with an evidence lattice applied to a robust selector, not the elementary lifting step. A prototype (10 nodes, 3 Byzantine) passes 16/16 falsification checks: byte-identical roots under adversarial gossip, deterministic re-convergence after late equivocation proofs, partition recovery, and three byte-identity-breaking ablations. The guarantee is *consistency*, not accuracy; robustness is imported, conditional on $\geq 2f + 3$ admitted contributions (at most f Byzantine) and a stated quantisation-margin condition.

1 Introduction

Robust aggregation and decentralised replication have developed on opposite sides of an architectural assumption. Byzantine-robust learning rules (Krum [5], coordinate-wise trimmed mean [41], Bulyan [32], FLTrust [10]) filter adversarial contributions at a central server. Decentralised training removes the server but replicates only the *training signal*, not the *filter*. Gossip-based SGD [30, 28] and periodic-sync schemes [15] yield replicas whose parameters agree statistically, not identically, and Byzantine variants of these schemes leave honest replicas free to exclude *different* nodes and therefore to compute *different* models. Systems that do force agreement on the filtered result do so by importing consensus, a total order from a blockchain or a BFT protocol, as in Krum-on-a-blockchain designs [18], and with it the coordination cost the decentralisation was meant to remove.

This paper shows the assumption is unnecessary for safety. A robust aggregation rule does not need an agreed order of arrival; it needs an agreed *set* of contributions and an agreed *exclusion predicate* over identities. Sets converge without consensus, which is precisely the guarantee of a state-based CRDT [37]. Exclusion-by-proof also converges without consensus, because a proof of equivocation (two signatures by the same key on different contents for the same round) is *self-authenticating*, so any party can verify it offline, its validity is objective, and the set of valid proofs only grows [13, 1]. Everything else in the pipeline is a pure function, and a pure function of converged state needs no coordination at all.

Why insist on byte-identity rather than a numerical tolerance? In a coordinated deployment a server anchors the truth and statistical drift is re-anchored every round. In a coordinator-free topology nothing re-anchors, so drift, however small per round, can compound across rounds, silently forking downstream state and foreclosing deterministic audit (recomputing exactly what a replica must have computed). Exact agreement closes both, and for selector rules the choice is not even available, because a selection flip moves the aggregate by a fixed quantum regardless of how small the perturbation was (Section 3.2), so no tolerance threshold is sound.

The construction, ACFA, extends the two-layer architecture of our prior work [19], which made non-associative model-merging operators convergent by separating state management (an OR-Set CRDT) from deterministic strategy execution. That paper’s discussion (§7.2, L4) sketched a trust-gated extension and left open whether the pattern “can deliver consensus-free Byzantine isolation in practice”. This paper answers that question affirmatively for a specific,

established class of rules, with three changes that the adversarial setting forces. The replicated state becomes a *product* of two lattices (contributions and evidence); the reduction becomes a globally coupled, discontinuous selector rather than a smooth merge (and the theorem admits stochastic selectors, though the reference rule is deterministic); the determinism obligation moves from an assumption on the execution environment (Assumption 10 of [19]) to a property of the arithmetic itself, because for discontinuous selectors no tolerance-based relaxation of that assumption exists (Section 3.2). The same discontinuity is why this is not an incremental patch to smooth trust-weighted designs such as the author’s E4 protocol [20]. There, quantisation tolerance absorbs rounding drift precisely because the aggregator is smooth; a selector has no such slack, and the arithmetic discipline plus the strict separation of the evidence lattice from the data lattice become forced rather than stylistic (Section 7).

Contributions.

1. A *product-lifting* theorem with a converse: any pure function of a converged product of CvRDTs inherits Strong Eventual Consistency, even when the function is non-monotone, non-associative, and stochastic; and an order-and-entropy-consuming implementation computes a well-defined function of the state *if and only if* it is invariant to input permutation and consumes only state-derived entropy (Section 5).
2. The ACFA construction: a contribution OR-Set composed with an equivocation-proof G-Set, an admission rule (per-round duplicate exclusion plus conviction), and a fixed-point full-dimension multi-Krum kernel resolved as a deterministic pure function of the converged product state (Section 4).
3. Accountability without votes, via the established exclusion-by-proof primitive [13, 1]: equivocation proofs are self-authenticating, conviction is monotone and permanent, and the part specific to this construction is that a proof arriving after an aggregate has been computed deterministically re-converges every honest replica to the same post-eviction output of the discontinuous selector (Theorem 10).
4. A prototype and falsification battery: 16/16 checks pass, including byte-identical roots under adversarial gossip, partition healing, late-proof re-convergence, and three ablations that each break byte-identity (Section 6).

What this paper does not claim. The guarantee is *consistency*, not accuracy. The replicated aggregate is exactly what the fixed-point robust rule computes on the admitted set, no better. The guarantee is *safety*, not liveness. As in any eventually consistent system, an adversary controlling message delivery can delay convergence indefinitely; the theorems condition on delivery, never on timing. The construction assumes a PKI, which restricts it to *permissioned* settings (cross-organisation consortia, regulated deployments); Sybil resistance, key rotation, and re-admission are delegated to it. Conviction is of the *key*. An adversary who steals an honest node’s key and equivocates with it evicts that identity *permanently*, with no revocation path here, so key theft is strictly worse than under per-round filtering, and re-admission under a rotated key is a PKI concern. Provable eviction covers exactly the violations that are self-authenticating (equivocation, which *convicts* the identity and evicts it permanently; malformed signatures are merely *excluded* by verification, without conviction); an adversary submitting plausible but poisoned values within protocol is bounded statistically by the robust rule’s own guarantees [5, 41], not provably. Finally, the consensus-free property is a statement about the *per-round resolve over a converged set*, not about a multi-round run: chaining round $r+1$ onto round r ’s aggregate requires each node to decide round r is *closed*, and that cut is a quiescence-detection (liveness) question a CRDT alone does not answer; nodes that cut at different points fork the round-DAG until they reconcile, without threatening the per-round safety guarantee. No total order is constructed, and none is needed.

2 Background

2.1 State-Based CRDTs and the Two-Layer Architecture

A convergent replicated data type (CvRDT) [37] is a state space with a merge operation that is commutative, associative, and idempotent; replicas that have received the same set of updates converge to identical states regardless of message order, duplication, or delay; this is eventual consistency [39] strengthened to Strong Eventual Consistency (SEC) [37]. The OR-Set supports add and remove with add-wins semantics [36]; a G-Set is the grow-only special case. Merkle trees over content-addressed elements provide integrity and a canonical commitment [35].

Our prior work [19] showed that neural-network merge operators fail the CRDT axioms structurally (normalisation, projection, and thresholding each break associativity), and made *any* such operator convergent with a two-layer architecture: Layer 1 replicates the contribution set as an OR-Set; Layer 2 applies the operator as a deterministic pure function over the canonically-ordered set, with randomness seeded from the Merkle root. Convergence of the

resolved value is conditional on strategy purity and on bitwise-deterministic execution (Assumptions 9–10 there). This paper reuses Layer 1 verbatim, replaces the assumption of environmental determinism with integer arithmetic, and adds a second replicated lattice for adversarial evidence.

2.2 Byzantine-Robust Aggregation

Given n contributed vectors of which at most f are adversarial, Krum [5] scores each vector by the sum of the squared distances to its $n - f - 2$ nearest neighbours and selects the minimiser; multi-Krum instead selects the m lowest-scoring vectors and averages them, where the selection count $m \in \{1, \dots, n\}$ is a free parameter ($m=1$ recovers Krum, $m=n$ plain averaging). This work fixes $m = n - f - 2$ (Definition 4). Coordinate-wise trimmed mean [41] discards the τ largest and smallest values per coordinate before averaging. Both come with statistical guarantees against $f < n/2$ -bounded adversaries under stated conditions ($n \geq 2f + 3$ for Krum; $\tau \geq f$ for the trimmed mean), and both are defined and analysed at a coordinator that sees all n contributions.

3 The Problem: Why Robust Selectors Resist Decentralised Replication

Three obstructions block naive decentralisation, each independently. Two are properties of the rules themselves, global coupling with non-foldability (§3.1) and discontinuity (§3.2), and one is a property of the adversarial setting, where equivocation forces an agreed exclusion predicate (§3.3).

3.1 Global Coupling and Non-Foldability

Krum’s score for contribution i depends on the pairwise distances among *all* contributions. Consequently the selection alone is not a sufficient running state: it cannot be recomputed incrementally from a partial selection plus new arrivals.

Lemma 1 (Non-foldability of multi-Krum). *There exist inputs on which multi-Krum selection over a set differs from multi-Krum applied to the multi-Krum selection of a subset together with the remaining elements. In particular, the natural fold (maintaining a partial selection and re-selecting on new arrivals) does not compute the direct selection.*

Proof. By explicit instance. For six 200-dimensional vectors drawn from a seeded Gaussian generator (seed 42; the generating script is included with the prototype), multi-Krum with $f = 1$ over all six selects indices $\{1, 3, 5\}$, while first selecting over the initial five and then applying multi-Krum to the selected subset augmented with the sixth yields a different selection of the *same size*. This same-cardinality comparison is the demanding one, and it exists only in this small- n regime: a re-selection over the $n - f - 2$ inputs to the second stage (the $n - f - 3$ survivors of the first stage plus the new arrival) has $m' = (n - f - 2) - f - 2 = n - 2f - 4 \leq 0$ exactly when $n \leq 2f + 4$, so the select-all convention returns exactly those $n - f - 2$ vectors, the direct selection’s count, but a different *set*. For larger n the fold cannot even match the cardinality; the fallback regime is thus where a fold could most plausibly reconstruct the direct selection, and it still does not. The transcript is reproducible from the artifact (Appendix A). $\square$

This is the analogue, for selection rules, of the associativity failure that motivated the two-layer architecture for merge operators [19]: an operator that cannot be folded cannot be maintained as running state on replicas that see contributions in different orders. The remedy there, deferring the operator until the set has converged, applies here as well, but the adversarial setting adds the two further obstructions below, which the original architecture does not address.

3.2 Discontinuity Closes the Tolerance Route

Replicas whose floating-point arithmetic differs by rounding cannot in general produce bitwise-identical outputs; our prior work handled this with an environmental determinism assumption (identical binaries, hardware, and rounding modes). One alternative is *tolerance*: for a smooth merge operator, replicas could agree within an ε rather than exactly. For selection rules, no such tolerance exists.

Lemma 2 (Selection discontinuity). *The map from contributions to the multi-Krum selection is discontinuous: at a score tie there exist configurations at which an arbitrarily small perturbation of one contribution changes the selected subset, and the two aggregates then differ by Δ/m , where $\Delta > 0$ is the gap between the swapped candidate vectors, an amount that does not vanish as the perturbation vanishes.*

Proof. Scores are continuous in the inputs, but the selection is an argmin over scores: on the boundary where two scores are equal, the selected subset changes discretely. When the two candidate vectors differ by Δ , the two aggregates (averages over subsets differing in one element) differ by Δ/m regardless of how small the score perturbation was. For instance, with $m = 5$ and two candidate vectors differing by $\Delta = 1$ in a single coordinate, a boundary flip shifts that coordinate of the aggregate by $1/5 = 0.2$, a jump no input-level tolerance can bound. $\square$

The consequence is that an input-level tolerance yields no output-level tolerance: replicas whose score computations differ by one unit in the last place can select different subsets and output aggregates separated by Δ/m (Lemma 2), independent of the perturbation. Approximate agreement protocols [14] recover bounded disagreement interactively, at the cost of rounds, the coordination this construction is designed to avoid. The remaining route is exact arithmetic: fixed-point integer computation, in which every operation is exact and identical on any architecture that implements integer semantics correctly. In this construction, integer arithmetic is not an optimisation preference; it is forced by Lemma 2. Section 6.4 exhibits the failure empirically: a floating-point variant of the kernel, differing only in accumulation order, produces divergent resolve transcripts.

3.3 Exclusion Must Be Agreed

A robust rule tolerates f adversarial *values*. It does not by itself handle an adversary that presents *different* values to different replicas (equivocation): replicas filtering locally may then exclude different identities, and their outputs diverge silently even when both are individually “robust”. Per-round statistical filtering also forgets: an identity excluded by the geometry in round r participates again in round $r + 1$. Agreement on *who is excluded* is therefore as necessary as agreement on the contribution set, and reaching it by voting would reintroduce consensus.

4 The Solution: Accountable Consensus-Free Aggregation

4.1 Architecture Overview

ACFA replicates two components and computes everything else:

- **Contribution OR-Set $\mathcal{C}$:** entries $\langle r, i, x, \sigma_i(r \| H(x)) \rangle$ (round, node identifier, fixed-point tensor, signature), content-addressed by hash. Merge is OR-Set union [36, 19].
- **Equivocation-proof G-Set $\mathcal{E}$:** entries $\pi = \langle r, i, H(x), H(x'), \sigma_i(r \| H(x)), \sigma_i(r \| H(x')) \rangle$ with $H(x) \neq H(x')$. Merge is set union. A proof is *self-authenticating*: its validity is decided by two signature verifications against the PKI, by any party, offline. Invalid entries are ignored by a validity filter inside the resolve function, so $\mathcal{E}$ need not be trusted as a set; only its valid subset matters, and validity is objective.

The replicated state is the product $(\mathcal{C}, \mathcal{E})$ with componentwise merge. The aggregation rule runs as a deterministic pure function of this product (Definition 4); by Theorem 7, its output inherits SEC.

4.2 Admission

Definition 3 (Admitted set). *Let $\text{Convicted}(\mathcal{E}) = \{i : \exists \text{ valid } \pi \in \mathcal{E} \text{ naming } i\}$. For state $(\mathcal{C}, \mathcal{E})$ and round r ,*

$$\text{Admit}(\mathcal{C}, \mathcal{E}, r) = \{e \in \text{Visible}(\mathcal{C}) : e.r = r, e.i \notin \text{Convicted}(\mathcal{E}), e \text{ is } e.i\text{'s unique visible round-}r \text{ entry}\}.$$

Here $\text{Visible}(\mathcal{C})$ denotes the OR-Set’s observable elements (added and not removed); contributions are never removed, so this is simply the element set, and the operator is kept for conformity with the OR-Set interface.

The uniqueness clause excludes same-round duplicates deterministically as soon as both are visible, so an equivocation never yields two admissible entries even before a proof circulates; the conviction clause makes the exclusion permanent across all rounds once a proof does. Entries with invalid signatures are likewise excluded inside the function. Honest replicas that observe both halves of an equivocation construct the proof themselves and add it to $\mathcal{E}$; proofs are canonicalised (the two hash–signature pairs ordered bytewise) so that all observers derive the identical proof object.

4.3 The Deterministic Kernel

Definition 4 (Resolve). Let $A = \text{Admit}(\mathcal{C}, \mathcal{E}, r)$, ordered by content hash (written sort_H). The resolved aggregate is

$$R(\mathcal{C}, \mathcal{E}, r) = \text{mean}(\text{MultiKrum}_f(A)),$$

i.e. the coordinate-wise mean of the $m = |A| - f - 2$ vectors selected by full-dimension multi-Krum, computed entirely in Q16.16 fixed-point (integer) arithmetic, with score ties broken by content hash. Here, if $|A| \leq f + 2$ (so $m \leq 0$) the selection degrades to all of A , and R of an empty admitted set is a distinguished bottom value, conventions that keep R total, as Theorem 7 requires and as the prototype implements.

Remark 5 (Design decisions forced by the adversarial review). An earlier kernel composed multi-Krum with a coordinate-wise trimmed mean over a 64-coordinate subsample seeded from $\text{PRF}(h_A \| r)$, where h_A is the Merkle root of the admitted set A . Adversarial review retired both: (i) a subsampled Krum is a different estimator than full-dimension Krum, so the imported robustness bound (Section 8) would not apply; and (ii) the subsample seed h_A is adversary-computable, so a last-mover Byzantine could grind its own vector against its own leaf $\rightarrow$ seed $\rightarrow$ coordinates to be selected on the sampled axes while poisoning the rest. Full-dimension multi-Krum restores the clean bound and removes this attack vector (no grinding vector was selected across 900 attempts, Section 6.3, X9). The Merkle-seeded PRF remains available for genuinely stochastic Layer-2 rules (Theorem 7 admits them), but the deterministic reference rule needs only the canonical hash tie-break. The rule is selectable: full-dimension multi-Krum by default, or Bulyan (a coordinate-wise median-trim after Krum selection, $|A| \geq 4f + 3$) where coordinate-concentrated robustness is required. The consistency guarantee (Theorem 14, Part A) is identical for either, since both are pure functions of the converged state, so the framework is robust-rule-agnostic and the consistency layer is orthogonal to the robustness rule.

Two choices carry the determinism obligation, and a third fixes the agreement scope. *Integer arithmetic*: every operation is exact, so the output is identical on any architecture implementing integer semantics under a fixed width contract (arbitrary-precision, as in the prototype, or a documented fixed width with overflow semantics); the environmental-determinism assumption of the prior architecture is thereby discharged rather than assumed. In the reference implementation all Q16.16 accumulation is performed in arbitrary-precision integers, precluding overflow; a fixed-width deployment must define and document its overflow semantics as part of the contract. *Canonical order*: multi-Krum is permutation-symmetric as a set function, so the hash order is load-bearing exactly at score ties and at entropy derivation, the two places where an implementation would otherwise consult arrival order or a local generator (both ablated in Section 6.2, A1/A2). *Admitted-set seeding*: entropy derives from a commitment to the admitted set rather than to the full replica state, so replicas agree on the output as soon as they agree on the admitted set, even while irrelevant state (invalid proofs, not-yet-delivered duplicates) still differs. Both the proof canonicalisation above and admitted-set seeding were forced by falsification runs against earlier versions of the construction that seeded from the full-state root (Section 6); each earlier version violated byte-identity in a late-proof scenario.

5 Formal Analysis

Theorem 6 (Product CvRDT). If $(S_1, \sqcup_1)$ and $(S_2, \sqcup_2)$ are CvRDTs, then $(S_1 \times S_2, \sqcup_1 \times \sqcup_2)$ is a CvRDT under the product order.

Proof. Commutativity, associativity, and idempotency hold componentwise; the product of join-semilattices is a join-semilattice with the componentwise join [36, 3]. $\square$

Theorem 7 (Product lifting). Let $S = S_1 \times \dots \times S_k$ be a product CvRDT and $F : S \rightarrow Y$ any pure total function. If two replicas have received the same set of updates (in any order, with any duplication) then F evaluates identically at both. F need not be monotone, associative, or continuous; stochastic F is admitted whenever its entropy is itself a function of the state (that nothing else is admissible is the converse, Theorem 8).

Proof. By Theorem 6 and the CvRDT convergence theorem [37], the two replicas' states are equal. A pure function evaluated at equal arguments yields equal values. A "stochastic" F whose randomness is drawn from a PRF keyed by the state is a pure function of the state. $\square$

Theorem 7 is a direct generalisation of Theorem 13 of the prior work [19], which is the case $k = 1$ with a deterministic strategy. The generalisation is elementary, and we present it as such; its value is the converse below, which delimits what an implementation may do, and the product structure, which lets adversarial evidence live in its own lattice, composed with the data lattice by the theorem. (The alternative, folding trust into a single shared lattice with the data, is realised by the author's E4 protocol [20]; the two designs are compared in Section 7.)

Theorem 8 (Characterisation of realisable implementations). *Let an implementation be a procedure P that consumes (a) some linearisation of the state and (b) an entropy source. P is replica-consistent (yields equal outputs on equal converged states) **if and only if** P is extensionally equal (almost everywhere in the entropy source; the measurability details are in the proof) to some $h(O(\text{state}), \text{seed}(\text{state}))$, where O is any state-canonical total order and seed is any state-derived function; equivalently, iff P is invariant under permutation of its input linearisation and consumes only state-derived entropy.*

Proof. ($\Leftarrow$) Immediate: both arguments of h are functions of the state. ($\Rightarrow$) If P is replica-consistent, its output depends on the state only through permutation-invariant features, hence equals a function of the multiset, and therefore of $O(\text{state})$ for any fixed state-canonical O . For the entropy clause, model the external source as a random variable Z and say P depends on Z if $z \mapsto P(\cdot, z)$ is not almost-surely constant under the law of Z . The output space Y is standard Borel (outputs are finite bit-strings or fixed-point tensors), so non-almost-sure-constancy is equivalent to the existence of a Borel output set C with $p := \Pr[P(\cdot, Z) \in C] \in (0, 1)$; put $A_C = \{z : P(\cdot, z) \in C\}$ and $B_C = \{z : P(\cdot, z) \notin C\}$, both of positive probability. (This routes through a measurable set rather than a point mass, so it holds even when the output law is atomless.) Two replicas with equal state and equal linearisation output $P(\cdot, Z_1)$ and $P(\cdot, Z_2)$ with Z_1, Z_2 i.i.d. copies of Z under the product law; if $Z_1 \in A_C$ and $Z_2 \in B_C$ then $P(\cdot, Z_1) \in C \not\equiv P(\cdot, Z_2)$, so independence gives $\Pr[P(\cdot, Z_1) \neq P(\cdot, Z_2)] \geq \Pr[Z_1 \in A_C] \Pr[Z_2 \in B_C] = p(1 - p) > 0$, contradicting replica-consistency. Hence P may consume only state-derived entropy. $\square$

Remark 9 (Why extensional, not syntactic). *The stronger-looking claim “ P factors as $h(\text{sort}_H, \text{seed})$ ” is false: a commutative fold (sum, count, maximum) is replica-consistent yet never applies sort_H and consumes no seed. Being a well-defined function of the state does not require syntactically consulting a canonical order. The reference kernel does use sort_H , but any state-canonical order serves; h absorbs the deterministic re-indexing.*

Theorem 10 (Accountable eviction). *Assume signed messages under a PKI and eventual pairwise delivery among honest replicas. Then: (i) validity of a proof is objective, so $\text{Convicted}(\mathcal{E})$ is equal at any two replicas with equal $\mathcal{E}$ and is monotone non-decreasing along any execution; (ii) once any honest replica holds a valid proof naming i , eventually all do, and every subsequent resolve at every honest replica evicts i , permanently; (iii) a proof arriving after a replica has already resolved changes its admitted set deterministically, and by Theorem 7 all honest replicas re-resolve to the same post-eviction output. No vote, quorum, or round is involved: the proof itself is the verdict, the accountable-safety / slashing pattern of two conflicting signed messages by one key [8], here carried, as in the blocklace [1], over a CRDT rather than a consensus vote, and gating a discontinuous selector’s admission rather than protecting the state.*

Proof. (i) Validity is two signature verifications against the PKI; a set union never removes a valid proof. (ii) Eventual delivery propagates the proof; conviction is a function of the valid subset of $\mathcal{E}$, which only grows. (iii) Admit and R are pure functions of the product state; Theorem 7 applies to the post-delivery states. $\square$

Remark 11 (Proof formation under partition). *Clause (ii) conditions on eventual delivery of an existing proof; forming one additionally requires that some honest replica observe both conflicting messages. Under a partition that perfectly separates the two halves of an equivocation from every honest node, no proof forms until the partition heals, so the two sides admit different entries and hold divergent roots for the duration. This interim divergence is the same class exercised separately by X3.b (late-proof lag) and X4 (partition healing); the combined equivocation-split-by-partition case is argued, not separately tested. Healing makes both entries visible everywhere, at which point the per-round uniqueness clause excludes the pair and a proof forms. Undetectability under perfect isolation is inherent to any asynchronous system without a dissemination commitment; it delays, but never subverts, conviction.*

Lemma 12 (Quantisation margin, a checkable no-flip condition). *Let contributions be quantised to a grid of step δ (for Q16.16, $\delta = 2^{-16}$), so each coordinate of the d -dimensional tensors carries error at most $\delta/2$. For admitted vectors the perturbation of a squared distance is bounded by $|\Delta\|x_i - x_j\|^2| \leq 2\delta\|x_i - x_j\|_1 + d\delta^2 =: \Delta^*$ (taking Δ^* with the largest pairwise L_1 distance; because this $L_1^{\max}$ is adversary-influenceable, an inflated contribution can enlarge β and deny certification (pushing a configuration into the empirical tier of Remark 13) but cannot produce a false certificate, since the bound is an upper bound on the true perturbation). Since a Krum score is a sum of the $|A| - f - 2$ smallest such distances (this $|A| - f - 2$ is Krum’s nearest-neighbour count, fixed by the score definition, not the selection count m of Definition 4), each of its $|A| - f - 2$ summands moves by at most Δ^* under the per-entry perturbation, so each score moves by at most $\beta := (|A| - f - 2)\Delta^*$. Hence if the boundary margin $g = s_{(m+1)} - s_{(m)}$ (the gap between the m -th and $(m+1)$ -th smallest real scores) satisfies*

$$g > 2\beta = 2(|A| - f - 2)(2\delta L_1^{\max} + d\delta^2),$$

the quantised multi-Krum selection **equals** the real-valued selection, and the imported robustness envelope holds exactly, up to the $\leq \delta/2$ per-coordinate value quantisation of the selected vectors plus one unit ($< \delta$) of floor rounding in the fixed-point mean. A replica evaluates $\hat{g}$ and $\hat{L}_1^{\max}$ on the quantised values it holds; since quantisation can shrink an L_1 distance by at most $d\delta$, the observable $\hat{\Delta}^* := 2\delta\hat{L}_1^{\max} + 3d\delta^2$ upper-bounds the real Δ^* , so $\hat{\beta} := (|A| - f - 2)\hat{\Delta}^* \geq \beta$; each score then moves by at most $\hat{\beta}$, so the observable gap obeys $g \geq \hat{g} - 2\hat{\beta}$, and checking $\hat{g} > 4\hat{\beta}$ on quantised data soundly certifies $g > 2\beta$. The observable form is conservative in two ways: the threshold doubles ($4\hat{\beta}$ versus 2β , from bounding $|g - \hat{g}| \leq 2\hat{\beta}$), which dominates, and $\hat{\Delta}^*$ inflates Δ^* by $\lesssim 4d\delta^2 \approx 2 \times 10^{-7}$ per squared distance. Even the fully-inflated score-level slack $2(|A| - f - 2)(\hat{\Delta}^* - \Delta^*) \approx 1.9 \times 10^{-6}$ stays below the minimum observed gap ($\approx 10^{-5}$), and the reported coverage uses the ideal real-score condition $g > 2\beta$, so nothing empirical changes. The coverage of Section 6.3 reports the real-score condition $g > 2\beta$ (measured against a finer-precision reference), which this observable form conservatively implies.

Proof. Expand $(a_k + \varepsilon_k)^2 - a_k^2 = 2a_k\varepsilon_k + \varepsilon_k^2$ with $a_k = x_{i,k} - x_{j,k}$, $|\varepsilon_k| \leq \delta$, and sum over coordinates for Δ^* ; each of the $|A| - f - 2$ summands of the sum-of- $(|A| - f - 2)$ -smallest moves by at most Δ^* under bounded per-entry perturbation, giving $|\Delta s_i| \leq (|A| - f - 2)\Delta^* = \beta$; two scores separated by g cannot reorder while each moves by at most β if $g > 2\beta$. $\square$

Remark 13 (The margin condition in three tiers). *Lemma 12 upgrades the qualifier of Theorem 14, Part B, from a hand-wave to a checkable inequality, and empirically (Section 6.3, 1600 configurations including deliberately near-tie ones at Q16.16) the quantised selection never flipped relative to a finer-precision reference. The condition resolves in three tiers. (i) Deterministic: the worst-case bound $g > 2\beta$ holds in 95.5–100% of configurations with a median safety factor of 11.5–346, guaranteeing no flip there. (ii) High-probability: an expected-error β (smaller by $\Theta(\sqrt{d})$, since the typical per-coordinate perturbation is far below the worst case $\delta/2$) raises coverage to 99.25–100% (an empirical coverage, not a proved tail bound; because the score perturbation is a sum of d bounded independent terms, a Bernstein/sub-exponential tail bound is derivable but we leave it open). (iii) Uncovered remainder: the $\lesssim 0.75\%$ left over are the $g \approx 0$ exact-tie configurations that no margin condition can cover ($g \rightarrow 0$). Under a continuous honest input distribution exact ties are a null event, so this tier is adversarial-only. We do not argue these are harmless, since a score tie need not mean the two candidate vectors are close (Lemma 2 shows a flip can move the aggregate by Δ/m); we report only that no flip occurred in any of the 1600 configurations, including the tight-scale ones, where the minimum observed boundary gap falls to $\approx 1.0 \times 10^{-5}$ (emitted by the sweep). (The near-duplicate-value bucket, by construction, drives its twin vectors deep inside the selection rather than onto its boundary; small g arises from small scale, not from the pairing.) The formally open item is a closed-form anti-concentration bound on the score-gap distribution; what the measurement settles is the practical question, that flips did not occur on the tested configurations, not the worst case.*

Theorem 14 (Composed guarantee, split). *The consistency and robustness legs have different strengths and are stated separately.*

Part A. Consistency (consensus-free, given a PKI and delivery). *Under a PKI, at quiescence on any common update set, all honest replicas output byte-identical aggregates and output roots.*

Part B. Robustness (conditional, imported, and rule-dependent). *The robustness level is a property of the plugged-in Layer-2 rule, not of the framework, and the two available rules give different guarantees. With full-dimension multi-Krum (the default; requires that the admitted set contain at most f Byzantine contributions, with $|A| \geq 2f + 3$), each selected vector satisfies the Krum criterion and lies within Krum’s Euclidean-distance envelope [5], under that result’s statistical hypotheses (i.i.d. honest contributions with bounded variance); the reported aggregate is their coordinate-wise mean, which inherits the envelope by convexity (the mean of points in a Euclidean ball lies in the ball). Blanchard et al. [5] state their tight (α, f) -resilience for the single Krum selection ($m=1$); its extension to the averaged multi-Krum ($m=|A| - f - 2$) is that work’s standard variant, imported here rather than re-derived. That envelope is Krum-level; Mhamdi et al. [32] showed it admits a bounded $O(\sqrt{d})$ coordinate bias under a coordinate-concentrated attack, where an adversary that places its whole budget on one coordinate while staying within the honest Euclidean spread is selected by Krum, and the plain mean carries the bias (Section 6.3, X5c, measures it at 2.29–2.35 across the two regimes; a bounded bias within the envelope, not a violation). With Bulyan ([32]; requires $|A| \geq 4f + 3$), the coordinate-wise median-trim after selection restores coordinate robustness (also X5c). Either way the envelope holds only under the quantisation-margin condition of Lemma 12: the boundary score-margin g must exceed 2β , the maximum quantisation-induced score perturbation. Under $g > 2\beta$ the quantised selection equals the real one and the imported bound holds exactly; bounding the flip probability when g is random is the remaining open piece (Remark 13).*

Part C. Accountability. *Every equivocating identity is eventually evicted from all subsequent rounds at all honest replicas. Liveness is not claimed anywhere: all guarantees are on states given deliveries.*

Proof. Part A is Theorem 7 applied to R , whose arithmetic is integer and exact, so no environmental-determinism assumption is needed, and sort_H is computed locally, so no coordinator or agreement is involved. Part B imports the guarantee of [5] for full-dimension multi-Krum (which $|A| = 10, f = 3$ satisfies, $10 \geq 9$, since in the base scenario all 10 nodes contribute and none is convicted); we do not re-derive it. The margin condition is not a cosmetic qualifier; by Lemma 2 multi-Krum is discontinuous, so quantisation, a bounded perturbation, can flip the selection near a score tie, moving the aggregate by Δ/m , and the over-the-reals bound holds only when the honest margin dominates that perturbation. Bounding the flip probability under a margin distribution is stated open work (Section 7.1), not an additive ε . Part C is Theorem 10. Note the asymmetry, in that Part A needs neither the admitted-Byzantine bound nor $|A| \geq 2f + 3$; those are robustness conditions only, so the consensus-free claim attaches to Part A alone. $\square$

Remark 15 (Interim divergence is inherent). *SEC promises equal outputs on equal delivered sets. Before deliveries complete, when one replica has seen an equivocation proof and another has not, admitted sets differ and outputs differ. This window is not a defect of the construction but the definition of eventual consistency; the construction’s contribution is that the window closes deterministically, with no reconciliation protocol, upon delivery alone (Theorem 10(iii)). Section 6 reports this window rather than hiding it.*

6 Experimental Evaluation

6.1 Prototype and Method

The prototype implements the construction end to end: Ed25519 signatures, SHA-256 content addressing, Merkle commitments, the contribution OR-Set and proof G-Set, admission, and the fixed-point kernel, with all kernel arithmetic in arbitrary-precision integers (no floating point and no ambient randomness anywhere in the resolve path). A simulator delivers events to in-process replicas under adversarial schedules: independent per-replica permutations, duplication, and partial-then-complete delivery. Default parameters: $n = 10$ nodes, $f \leq 3$, 200-dimensional $Q16.16$ tensors. The reference kernel is full-dimension multi-Krum (Definition 4); it does no coordinate subsampling and, on its deterministic path, consumes no randomness beyond the content-hash tie-break. Test-generation randomness is seeded and the transcript is reproducible from one command; the kernel itself is entropy-free. The scale is deliberately small, chosen so that every check is exact and the pass criterion is binary: equality of 32-byte output roots. Cross-architecture identity is a property of integer semantics rather than a measurement, so a single machine suffices to falsify the remaining claims; we have not benchmarked performance, and none of the checks below depends on timing.

6.2 Falsification Battery

Table 1 summarises the 16 checks. All pass.

Three aspects deserve comment. First, the ablations are as informative as the positive checks: each removes exactly one mechanism (canonical tie-breaking, state-derived entropy, integer arithmetic) and each breaks byte-identity, which is what the design claims and what a design with redundant mechanisms would fail to show. Second, the late-proof check (X3.b) exercises Theorem 10(iii) in the adversarially interesting order: a replica commits an aggregate that includes the eventually-convicted identity, receives the proof afterwards, and re-converges to the other replicas’ root with no protocol beyond delivery. Third, earlier versions of the construction failed X3.b and X3.c: seeding the kernel from the full-state Merkle root let a garbage proof, correctly ignored by admission, still perturb the output through the seed. The falsifier drove the two design corrections reported in Section 4.3 (admitted-set seeding; canonical proof objects). We report this because the failures are informative about where the determinism obligations actually sit.

6.3 Scale, Robustness, and Cost Battery

A second battery (Table 2) tests what the falsification checks do not: order-invariance at larger scale, the robustness leg an adversary actually targets, the guarantee-class contrast, and the cost of quantisation.

Two points are worth stating plainly. The X6 delta is genuinely zero only because $Q16.16$ is lossless *for this task*; it is not a general claim that quantisation is free, and X5 shows the kernel tracks its float reference rather than being identical to it. And X9 is the test the adversarial review demanded: the earlier subsampled kernel was grindable, and only the full-dimension rule (Remark 5) withstands a last-mover search, but robustness remains an *imported* guarantee under the quantisation-margin condition of Theorem 14, Part B, not a contribution of this paper.

Table 1: Falsification battery. Pass criterion is byte-equality of output roots except where noted. Ablations are expected to *diverge*: they remove one mechanism each, and a pass means byte-identity was broken.

Check	Result	Content
X0.1 purity	pass	same state resolved twice $\rightarrow$ identical bytes
X0.2 canonicity	pass	200 arrival permutations of one state $\rightarrow$ one root
X0.3 non-foldability	pass	explicit instance for Lemma 1
X2 order-invariance	pass	20 adversarial schedules $\times$ 3 rounds $\times$ 5 replicas, with duplication and partial-then-complete delivery $\rightarrow$ all roots byte-identical
X3.a duplicate exclusion	pass	both halves of an equivocation visible $\rightarrow$ deduplicated identically at all replicas
X3.a proof formation	pass	observers of both halves auto-form the canonical proof
X3.b interim window	pass	replica without the proof diverges (Remark 15), by construction of the test
X3.b late proof	pass	proof delivered after resolve $\rightarrow$ all replicas re-converge to the same root
X3.b permanence	pass	convicted identity excluded in a later round
X3.c proof spam	pass	invalid proof delivered $\rightarrow$ output unchanged
X4 partition (isolated)	pass	three partitions resolve to three distinct roots while isolated
X4 partition (healed)	pass	after healing, all 10 replicas equal a fresh replay’s root
A1 tie-break ablation	diverges	arrival-index tie-breaking on a tied-score instance breaks byte-identity
A2 entropy ablation	diverges	a stochastic Layer-2 variant (seeded coordinate sub-sample) driven by a node-local generator instead of the state-derived seed breaks byte-identity
A3 arithmetic ablation	diverges	floating-point kernel, two accumulation orders, breaks the resolve transcript
X0.4 duplicate signature	pass	a malleated second signature over the same tensor is either rejected by canonical verification or excluded by the uniqueness clause; roots agree under both arrival orders (defence in depth for Definition 3)

The quantisation-margin measurement. To test Lemma 12 we swept 1600 configurations (honest random and near-duplicate-value, each at two weight scales; fixed RNG seed 3, and the qualitative three-tier resolution (Remark 13) is the claim, the exact band is seed-conditional), comparing the $Q16.16$ multi-Krum selection to a finer-precision reference and recording the boundary margin g against the worst-case perturbation 2β . There were **zero selection flips** in all 1600; the sufficient condition $g > 2\beta$ held in 95.5–100% of configurations with a median safety factor of 11.5 to 346 (a tighter expected-error β raises this to 99.25–100%), and even the residual near-tie configurations, where the worst-case β leaves the condition inconclusive, flipped nothing. We do not claim the uncovered tail is harmless; by Lemma 2 a flip there would move the aggregate by Δ/m , we report only that none occurred across the 1600 configurations, including the tight-scale bucket whose minimum observed boundary gap is $\approx 1.0 \times 10^{-5}$, the genuine boundary probe (Remark 13).

6.4 Observations on the Determinism Envelope

Two incidental measurements from constructing ablation A3 bear on how floating-point non-determinism manifests in practice. CPython 3.12+ implements the built-in `sum` with compensated (Neumaier) summation [33, 34], so a naive-versus-`fsum` comparison shows no divergence; the divergence appears in plain loop accumulation, the order-dependent pattern of C and BLAS kernels [21, 24], where 39 of the 45 pairwise distance sums differ between two accumulation orders in the artifact instance (the count is emitted by the battery itself). Separately, values quantised to $Q16.16$ and then converted to doubles sum *exactly* in floating point regardless of order, because all terms lie on a common dyadic grid with every partial sum’s numerator below 2^{53} (comfortably satisfied at this scale); quantisation alone therefore buys order-invariance for sums, though not for the full nonlinear kernel. Neither observation substitutes for the integer kernel; both delimit where the hazard lives. $Q16.16$ also fixes the dynamic range at $\pm 2^{15}$ with 2^{-16} resolution; gradient components far below that resolution quantise to zero, so the format is a deployment parameter, not a universal choice. Any per-tensor scaling introduced to widen the range must itself be a deterministic state-derived function; otherwise it reopens the exact non-determinism the integer kernel exists to close.

7 Discussion

Relation to the two-layer architecture. ACFA is the L4 extension sketched in [19] made concrete. Where that architecture assumed environmental determinism (its Assumption 10), the present construction discharges the obligation with integer arithmetic, which Lemma 2 shows is not optional for selection rules.

Relation to E4, and the reduction-class distinction. A closer relative is the author’s own E4 protocol (`crdt-merge` v0.9.5+; [20]), which realises the single-shared-lattice form of that sketch: a multi-dimensional trust lattice in which trust propagates as CRDT data through the same pipeline it validates. E4 already provides several primitives this paper also uses, namely cryptographically self-authenticating equivocation evidence (two conflicting signed operations from one peer at the same sequence number), a canonical-order deferral of an order-dependent resolver, and a fixed-point integer accumulation mode, so E4 is prior work this paper builds on, not around, and the distinction is precise. It lies in the *reduction class* and the *actuation*, not in the use of proofs. E4’s resolver is a *trust-weighted mean with outlier detection*, a smooth, order-independent aggregate. For a smooth aggregate, deterministic quantisation of the parameters, and full-precision re-anchoring when accumulated rounding drifts past a threshold, suffice for convergence by the *tolerance* route. ACFA’s reduction is instead a distance-geometry *selector* (multi-Krum), which Lemma 2 shows is *discontinuous*. A bounded perturbation can flip the selection and move the aggregate by Δ/m . For such a selector the tolerance route is closed: no quantise-and-re-anchor scheme can guarantee agreement, because the disagreement is not a bounded numeric drift but a discrete selection flip. This is why ACFA discharges determinism with *exact integer arithmetic* rather than quantisation-with-re-anchoring, a different mechanism forced by the selection class, not a refinement of the same one. Likewise, where E4 converts equivocation evidence into a *graded, progressive* trust reduction that feeds the trust-weighted mean, ACFA converts the same evidentiary kind of proof into a *binary, permanent* conviction that gates admission, a different effect suited to a selector that has no notion of a soft weight. The evidentiary primitive is shared; the aggregation it feeds, and the arithmetic that aggregation forces, are what differ.

Multi-round execution and epoch forking. The guarantee is per-round. Deciding that round r is *closed* is quiescence detection (a liveness question a CRDT alone does not answer), and replicas that cut at different points fork the round-DAG until they reconcile (Section 1): one replica may train on round- r aggregate Y_r while another, having admitted a late proof, has re-resolved to Y'_r and moved on. Per-round safety is unaffected by the cut choice (any two replicas that close r on the same contribution set compute the same bytes), but multi-round *liveness* requires an orthogonal epoch-synchronisation mechanism (synchronous timeouts, threshold signatures on round closure, or any transport-layer commitment) which this paper explicitly delegates. Such a mechanism agrees only on *when* a round closes, never on the aggregate *value*, which stays a consensus-free pure function of the admitted set; the coordination removed is agreement on the computation, and only round-closure timing is delegated.

The adaptive margin adversary. Lemma 12 is a worst-case condition against *static* inputs; an adaptive last-mover that observes honest contributions could compute the exact integer scores and craft a vector forcing $g \approx 0$, landing the configuration in the uncovered tier of Remark 13. Two structural features mitigate this, though neither closes it. First, for a *single-shot* adversary, ties are broken by content hash, a state-canonical order it cannot steer independently of its own vector’s bytes, so forcing a tie costs it control over *which* side wins, degrading a targeted flip to an untargated Δ/m perturbation, which is what the imported envelope already prices in. A *grinding* last-mover, however, may search jointly over the tie constraint *and* its own resulting hash, and so is not excluded by this argument; that residual is exactly the item Remark 13 names as open (a closed-form anti-concentration bound on the score-gap distribution). Second, the distinct get-*admitted* grinding variant, a last-mover searching candidates to place itself in the selection, is measured and closed (X9: 0/900 selections).

Provable versus statistical exclusion. The construction separates two adversary classes cleanly. Equivocation is *convicted* provably and evicted *permanently*, with no vote; malformed signatures are excluded by verification without conviction; per-round duplicates are excluded by the uniqueness clause, likewise without conviction. Adversaries that stay within protocol and submit poisoned values are bounded *statistically* by the robust rule, per round. The composition is strictly stronger than per-round filtering alone (an equivocator is removed from all future rounds, not re-filtered each round), but it is not a claim that all Byzantine behaviour is provable: a within-bounds poisoner leaves no proof.

Costs. Relative to a coordinator, the costs are full-state or delta gossip of the contribution set, one signature per contribution and two verifications per proof; the fixed-point kernel (integer multi-Krum is $O(n^2d)$ [5] like its float counterpart); and quantisation of contributions to Q16.16. Relative to consensus-based decentralisation, there are no rounds, no quorums, and no leader; the price is the weaker guarantee, eventual, not immediate, agreement.

7.1 Limitations and Future Work

The robustness leg (Theorem 14, Part B) is the weakest part and the adversarial review sharpened it. (a) *The quantisation-margin condition is the real open problem:* because multi-Krum is discontinuous (Lemma 2), Q16.16 rounding can flip the selection near a score tie, moving the aggregate by Δ/m , so the imported over-the-reals bound holds only when the honest score margin dominates the quantisation perturbation; bounding the flip probability under a margin distribution is open, and is not the additive ε an earlier draft implied. (b) *Robustness needs an admitted-population condition* ($|A| \geq 2f + 3$ among admitted contributions), a delivery/completeness condition that the consistency leg (Part A) does not require; we separate the two so the consensus-free claim is not overstated. (c) *The default rule is Krum-level, not Bulyan-level:* full-dimension multi-Krum is vulnerable to coordinate-concentrated attacks [32] (Section 6.3, X5c); Bulyan restores coordinate robustness but needs $|A| \geq 4f + 3$, a stronger admitted-population condition than the $|A| \geq 2f + 3$ of the default. The consistency guarantee is unaffected by the choice. More fundamentally, Krum and Bulyan are themselves defeated by within-norm attacks that stay inside the honest spread [4, 38], exactly the stealth drift X5b reports as unfiltered. This is a property of the imported rule, not the framework; ACFA replicates whatever robust rule is plugged in, so a stronger successor to Krum inherits the same consistency and accountability guarantees unchanged. (d) *Cross-architecture byte-identity is argued, not measured:* the integer kernel is exact under a fixed width contract, but the prototype runs one architecture; a heterogeneous run (or a fixed-width port with documented overflow semantics) is owed. The prototype validates correctness, not performance: no timing, bandwidth, or large- n measurements are reported, and delta-state propagation [2] would be required at scale, as in the parent architecture. The equivocation G-Set grows monotonically; rate-limiting proof spam (all of it invalid by construction, but bandwidth-consuming) and garbage-collecting evidence for departed identities are engineering questions we have not addressed. Sybil resistance is delegated to the PKI. Finally, the accountable class here is equivocation; extending self-authenticating evidence to other violation classes (for example, provably malformed training provenance) is future work.

8 Related Work

Byzantine-robust aggregation, central and decentralised. Krum [5], trimmed mean [41], Bulyan [32], and FLTrust [10] define and analyse robust rules at a coordinator. This paper does not modify the rules or their bounds; it replicates their evaluation without the coordinator. The difficulty of doing so is known: applying geometry-based selectors in a decentralised topology makes honest nodes filter *different* peer sets and therefore disagree. Wu, Chen and Ling [40] name this “disagreement” precisely and address it with an iterative outlier scissor (IOS) scheme carrying optimisation-convergence guarantees; Cajaraville-Aboy et al. [9] (WFAgg) combine complementary distance-, cosine-, and temporal-based filters that individually miss different attacks, and propose a weighting aggregator; and BALANCE [17] gives a decentralised Byzantine-robust method with convergence guarantees. Closest in surface form is BRACE [16], which distributes one quantised aggregate through the ring’s lockstep reduce with a coordinate-wise *consensus* on the sign and a convergence guarantee; any agreement there is a product of the consensus this work removes, not of a pure function of converged state. We do not claim the problem is open. Our distinction is the *guarantee class*: those works achieve statistical or optimisation convergence in a neighbour-gossip topology, where honest replicas end near one another. ACFA instead achieves *exact byte-identical* agreement on the robust aggregate over a replicated contribution set, via deferred deterministic replay (the accountability engine of PeerReview [22], here discharged for a discontinuous selector by exact integer arithmetic) plus provable accountability. Consistency, not convergence rate, is the object.

Byzantine fault tolerance in CRDTs. Kleppmann [27] makes CRDT *state* tolerate Byzantine replicas via hash-graph techniques; the object protected is the replicated set itself. The blocklace [1] goes further, using self-authenticating signed-hash evidence to detect equivocation and eventually *exclude* equivocators inside a CRDT without consensus, the exclusion-by-proof primitive our accountability leg reuses; concurrently, Brocco [6] obtains Byzantine-resilient convergence by deferring a deterministic interpretation over accumulated CRDT state, and a concurrent follow-up [7] decouples identity-based from content-based trust for fine-grained post-compromise handling; our admission gate is deliberately coarser, a *binary, permanent* identity conviction, not a graded per-update

trust, because the discontinuous selector it gates admits no soft weight. The present work is complementary to all three: it assumes eventual pairwise delivery among honest replicas for state propagation (the $|A| \geq 2f + 3$ bound constrains the contribution population, not propagation) and protects a non-associative *reduction over* the state, gating that reduction’s admission by the exclusion proof, an object those constructions do not protect: the blocklace and Brocco’s layer defend convergence of the replicated state or of a benign interpretation of it, whereas the selector here is discontinuous in and adversarially coupled across its inputs. Where a subclass of CRDTs can simply *tolerate* equivocation and retain strong eventual consistency [25], our reduction cannot: two admitted entries from one equivocator would flip the discontinuous selector, so equivocation must be *excluded*, not absorbed. Accountable consensus [11] detects and proves misbehaviour *within* agreement protocols; we import the self-authenticating-evidence idea into a setting with no agreement protocol at all.

Decentralised training. Gossip SGD [30, 28] and low-communication schemes [15] target statistical convergence of training; replicas hold similar, not identical, parameters, and Byzantine handling (where present) is per-replica and unsynchronised. Federated learning [31, 26] centralises aggregation. The guarantee class here, bitwise agreement on the robust aggregate, coordinator-free, is different from both, and is the guarantee the state-convergence line [37, 19] provides once the reduction is made a pure function of converged state.

Composition of CRDTs with queries. That a deterministic query of a converged CRDT is replica-consistent is implicit in the OR-Set definition [36], which builds query-agreement into the datatype rather than stating it as a separate result; our prior work formalised it for one OR-Set and a deterministic strategy [19]. Which computations over a lattice are order-independent has a longer lineage. LVars restricts reads to monotone threshold queries [29], and the CALM line ties coordination-freedom to monotonicity [12, 23]; our converse (Theorem 8) instead admits *non-monotone* reductions by deferring to a converged (quiesced) set, characterising consistency by permutation-invariance plus state-derived entropy rather than by monotonicity. The present contribution is the product form with an evidence lattice, the explicit admission of non-monotone stochastic reductions with state-derived entropy, the converse (Theorem 8), and the demonstration that this suffices for an established class of Byzantine-robust rules.

9 Conclusion

Byzantine-robust aggregation rules are globally coupled, non-associative, and discontinuous, three properties that appear to demand a coordinator or a consensus layer. They do not. The rules need an agreed set and an agreed exclusion predicate; both are join-semilattices, both converge without coordination, and everything downstream is a pure function of their product, which Theorem 7 makes replica-consistent wholesale. The contribution is a composition and an application, not a new algebra: the product-lifting theorem is elementary (Theorem 7) and its value is the converse (Theorem 8) plus the demonstration that a discontinuous, globally-coupled robust selector can be made strong-eventually-consistent with exact arithmetic and accountable eviction. Consensus-free accountable eviction is itself established: self-healing lattice agreement evicts faulty replicas over a lattice via undeniable proofs [13], and the blocklace excludes equivocators inside a CRDT by self-authenticating evidence [1]. What we did not find in the Byzantine-CRDT or Byzantine-FL prior art is that primitive composed with a discontinuous, globally-coupled robust *selector* as the gated object, under the exact-arithmetic determinism obligation that discontinuity forces. The two obligations that remain, exact arithmetic, because discontinuity voids any tolerance (Lemma 2), and state-derived entropy, because the converse theorem permits nothing else (Theorem 8), are both dischargeable by construction, and the 16-check falsification battery, including three ablations that break byte-identity by removing one mechanism each (a fourth, A4, in the extended battery), locates the load-bearing points empirically. Equivocation is handled by evidence rather than by vote: a proof any party can check offline, accumulated in a grow-only lattice, evicting deterministically even when it arrives after the fact. The result is a coordinator-free replication discipline for exactly the aggregation rules that adversarial distributed learning already uses, with the scope limits stated in Section 1: consistency not accuracy, safety not liveness, and proofs only for what is provable.

Reproducibility. The prototype and all test harnesses will be released publicly upon publication and are available from the author on request in the interim. Every reported number is regenerated by a fully seeded script: `python3 falsifier.py` reproduces the 16-check transcript of Table 1 (and Appendix A); `python3 battery.py` the extended battery of Table 2 (rows X1, X5a, X5b, X6, X8, A4), with `python3 x9_sweep.py` regenerating the X9 grinding sweep (0/900; 300 trials) and `python3 x5c_bulyan.py` the X5c coordinate-attack row (2.29/2.35, Bulyan 0.00); and `python3 t5_flipprob.py` the quantisation-margin sweep of Section 6.3 (RNG seed 3).

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A Falsifier Transcript Summary

The battery runs as `python3 falsifier.py` (CPU only; about four seconds on commodity hardware). All test-generation seeds are fixed in the script (seed 42 for the battery; the A3 floating-point instance uses its own fixed seed); the kernel consumes no entropy beyond the state-derived content-hash tie-break. The Lemma 1 instance is emitted by check X0.3: with $f = 1$ over six seeded 200-dimensional Gaussian vectors, global multi-Krum selects indices $\{1, 3, 5\}$ while the fold path (select over the first five, then select over the selection plus the sixth) yields a different subset *of the same cardinality*, the fallback regime (second-stage $m = 0$) that Lemma 1 identifies as the demanding case for a fold. The A3 ablation instance uses unquantised Gaussian doubles: with plain loop accumulation, 39 of the 45 pairwise distance sums differ between arrival-order and sorted-order accumulation, and the hashed resolve transcripts diverge; the same comparison under CPython’s compensated built-in `sum`, or over $Q16.16$ -quantised doubles (a common dyadic grid), shows no divergence, as reported in Section 6.4.

Table 2: Extended battery. X-checks measure; the A4 ablation is expected to *diverge* (removing per-round duplicate exclusion lets an equivocator split admitted sets), and a pass for A4 means byte-identity was broken. All pass.

Check	Result	Content
X1 order-invariance at scale	pass	50 rounds $\times$ 7 replicas $\times$ 30 adversarial schedules with duplication $\rightarrow$ all roots byte-identical
X5a fixed-point tracks float	pass	full-dimension multi-Krum kernel matches its float64 reference to $\sim 10^{-4}$ across four attack types (the consistency arm of robustness)
X5b better than naive	pass	under large-norm / sign-flip / colluding-shift attacks the aggregate deviation from the honest mean stays far below the naive all-inclusive mean, a sanity contrast, <i>not</i> a verification of the imported Blanchard envelope (which Theorem 14, Part B imports rather than measures); stealth within-norm drift is not filtered, reported as a non-claim
X9 grinding Byzantine	pass	a last-mover adversary that searches 400 candidates to minimise its own full-dimension Krum score is selected in 0 of 900 attempts (300 trials $\times$ 3 grinders), and the aggregate stays within the honest-only Krum spread (the maximum deviation among honest-only Krum aggregates across seeds) in all 300 trials; the subsample removal (Remark 5) removes this attack vector
X5c coordinate attack ($n=16, f=3$)	pass	the Mhamdi coordinate-concentrated attack (whole budget on one axis, within the honest Euclidean spread) is <i>selected</i> by Krum, all 3 of 3 Byzantine vectors, in each of the two seeded regime instances (colluding / non-colluding): plain multi-Krum shows a coordinate deviation of 2.29 (colluding) / 2.35 (non-colluding), a <i>bounded</i> $O(\sqrt{d})$ Krum-level bias within the imported envelope, not a violation, while Bulyan’s per-coordinate median-trim drives it to 0.00 (in both colluding and non-colluding regimes), because the $\theta=n-2f$ selected are honest-majority per coordinate
X8 guarantee-class contrast	pass	on non-comparable objects (multi-round gossip parameters vs. a single replicated aggregate), gossip-averaging SGD leaves replicas ≈ 0.09 apart in L_∞ at quiescence while ACFA agrees to 0 bytes, a guarantee-class contrast (statistical vs. bitwise), not a performance comparison
X6 quantisation cost	pass	on one learnable task, fixed-point and float robust-aggregation training reach the same held-out accuracy (0.985, three significant figures, single seed); this bounds the Q16.16 quantisation cost below the metric resolution for that task only, not in general
A4 dedup ablation	diverges	removing the per-round uniqueness clause lets an equivocator split admitted sets between proof-lagged replicas $\rightarrow$ divergent roots